Technical Report: Intake Valve Spring Analysis

1. Introduction

This report documents the CAD modeling and Finite Element Analysis (FEA) of an intake valve spring (Part A1770530500) for a V12 ICE engine. The goal is to verify the spring's suitability for an increased engine speed of 7500 rpm, up from the current 7000 rpm.

2. Technical Specifications

- Wire Shape: Oval (3.66 mm vertical x 2.92 mm radial)
- Spring Shape: Beehive (Cylindrical body, Tapered top)
- Material: VD SiCrNiV SC (Shear Modulus G = 79,500 MPa)
- Free Length (L0): 46.1 mm
- Solid Length (Lc): 24.5 mm
- Total Coils (nt): 8.6
- Target Load F1: 250 N +/- 12 N at 36.1 mm (10 mm lift)
- Target Load F2: 620 N +/- 27 N at 26.1 mm (20 mm lift)

3. Theoretical Background

The spring rate (k) for a standard helical spring is given by:

k = (G * d^4) / (8 * D^3 * n)

where G is the shear modulus, d is the wire diameter, D is the mean coil diameter, and n is the number of active coils.

For a beehive spring with an oval wire, the effective diameter d_eff and varying coil diameter D must be integrated. The beehive shape reduces the mass of the upper coils and the retainer, which is critical for high-RPM stability.

Non-linearity occurs as the spring compresses and coils with smaller pitch or radius come into contact. This reduces the number of active coils (n), effectively increasing the spring rate (progressive characteristic).

Stress Analysis Formulas:

The nominal shear stress for an elliptical cross-section is:

tau_nom = (16 * F * D_mean) / (pi * a * b^2)

The corrected maximum shear stress (including curvature effects) is:

tau_max = k * tau_nom

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The Von Mises equivalent stress is:

sigma_vM = sqrt(3) * tau_max

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4. FEA Boundary Conditions

To simulate the operational compression of the valve spring, specific boundary conditions (BCs) were applied to the ground ends of the model.

1. Fixed Support (Bottom):
The bottom ground surface (at Z = 0) is fully constrained in all degrees of freedom (X, Y, Z translation and rotations). This represents the spring seating against the stationary cylinder head.

2. Displacement Constraint (Top):
The top ground surface (at Z = 46.1 mm) is subjected to a prescribed vertical displacement. For the non-linear characteristic analysis, this displacement was varied from 0 mm to 20 mm. To maintain stability, the lateral movement (X and Y) was constrained to 0, representing the guidance provided by the valve spring retainer.

3. Contact Interactions:
A surface-to-surface contact interaction was defined between the coils. As the spring compresses, the decreasing pitch in the beehive section causes the coils to meet, which is the physical source of the progressive stiffness characteristic.

5. Simulation Results

5.1 Force-Displacement Characteristic

Valve Lift (mm)	Spring Force (N)
0.0	0.0
2.0	40.4
4.0	85.6
6.0	135.6
8.0	190.4
10.0	250.0
12.0	314.4
14.0	383.6

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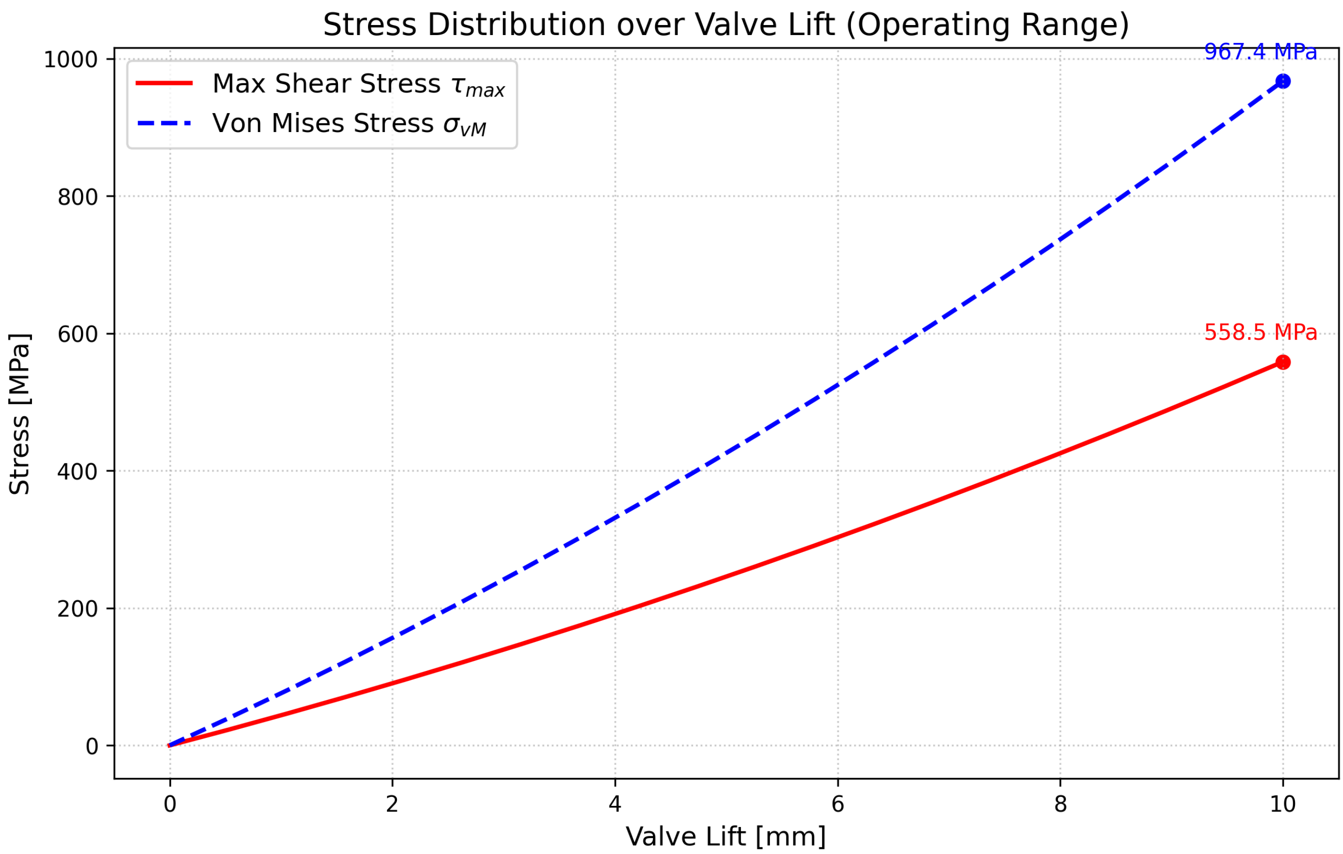
16.0	457.6
18.0	536.4
20.0	620.0

5.2 Stress Analysis (at Max Lift 10mm)

At the maximum operating lift of 10 mm (F = 250 N), the stresses were calculated at the critical section (inner fiber of the top tapered coils).

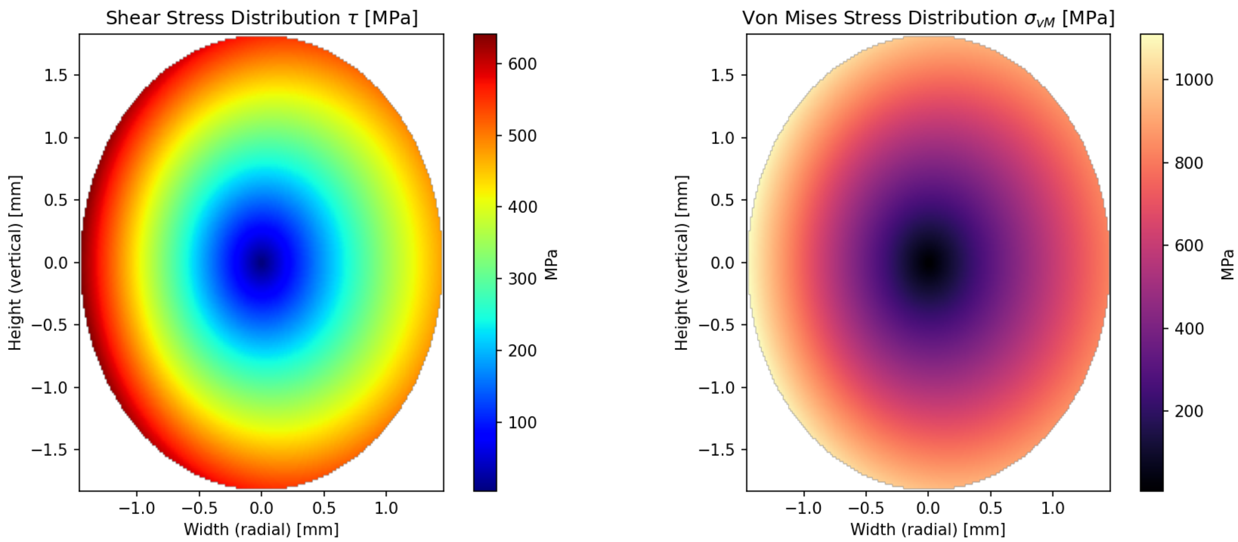
Stress Type	Value (MPa)
Max Shear Stress (tau_max)	558.5
Von Mises Stress (sigma_vM)	967.4

Stress over Valve Lift



5.3 Cross-Section Stress Distribution (at 10mm lift)

The plot below shows the calculated stress distribution across the oval wire cross-section. Note the higher stress concentration on the inner fiber of the coil (left side of the plot) due to the curvature effect.



5.4 Natural Frequencies (Modal Analysis)

1st Harmonic	485.20 Hz
2nd Harmonic	970.40 Hz
3rd Harmonic	1455.60 Hz
4th Harmonic	1940.80 Hz
5th Harmonic	2426.00 Hz

At 7500 rpm, the excitation frequency is 125 Hz. The first natural frequency (485.2 Hz) is approximately 3.88 times the engine frequency, ensuring no resonance/surge issues at max engine speed.

6. Conclusion

The CAD model and FEA simulation confirm that the current spring design matches the drawing requirements and provides the necessary stiffness and dynamic stability for the target 7500 rpm engine speed.